A Multimodal Image–Text Fusion Framework for Automated Skin Lesion Analysis, ABCDE Evaluation, and Interactive Medical Query Support Using a Patient-Centric Conversational System

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1. Project Overview

This project aims to assist in **skin lesion cancer assessment** by collecting and integrating **multimodal patient information** through a **patient-centric conversational system**. The system gathers the following inputs:

- **Skin lesion images** (clinical or dermoscopic)
- **Patient-reported medical symptoms and history**
- **Medical reports**, such as biopsy findings or blood test results (e.g., elevated LDH levels)
- **Lesion evolution information** provided by the patient, including increase in size, color changes, itching, bleeding, or spreading

All data is collected via a **chat-based interface**, where patient responses, images, and documents are processed continuously. The system maintains **chat history and contextual memory** to ensure consistent understanding across all stages.

The overall pipeline includes **image preprocessing, lesion segmentation, ROI extraction, hierarchical classification, medical text processing, multimodal image–text fusion with ABCDE evaluation, risk scoring, and Visual Question Answering (VQA)** to provide explainable diagnosis assistance.

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2. Stage-Wise Workflow

Stage 1: Preprocessing of Datasets

Purpose

To prepare clean and standardized datasets for training and inference across all downstream components.

Preprocessing Performed

- Image resizing and normalization
- Removal of noise and hair artifacts (if required)
- Mask–image pair alignment for segmentation datasets
- Text cleaning and normalization for medical descriptions
- Dataset splitting into training, validation, and testing sets

Input

- Public skin lesion datasets
- Patient medical datasets
- Collected medical reports

Output

Clean, standardized datasets ready for training **SLS, SLRC, HC, and MedProc** models.

Stage 2: Skin Lesion Segmentation (SLS)

Purpose

To isolate the lesion region from the full skin image.

Input

- Original high-resolution skin lesion image

Preprocessing Required

- Resize image to model resolution
- Normalize pixel values
- Optional contrast enhancement

Operations Performed

- Generate a binary lesion segmentation mask
- Remove surrounding non-lesion skin regions
- Extract lesion boundaries and contours

Output

- Binary segmentation mask
- Cropped binary lesion region
- Geometry information (contours, boundaries, pixel area)

Stage 3: Skin Lesion ROI Cropping (SLRC)

Purpose

To extract the exact lesion region in **original color** using the segmentation mask.

Input

- Original skin image
- Segmentation mask from SLS

Operations Performed

- Align segmentation mask with original image

- Crop Region of Interest (ROI) while preserving lesion borders
- Remove non-lesion background regions

Output

- High-quality color lesion ROI
- Updated bounding-box coordinates
- Clean image ready for classification

Stage 4: Hierarchical Classification (HC) – Updated

Purpose

To perform hierarchical lesion classification and extract **deep visual and geometric features** required for downstream clinical reasoning.

Input

- Color lesion ROI from SLRC
- Segmentation geometry from SLS

Preprocessing Required

- Image normalization
- Aspect-ratio correction

- Minor noise reduction (if required)

Operations Performed

- **Hierarchical lesion classification**
  - Benign vs malignant classification
  - Sub-category classification (melanoma, nevus, etc.)
- **Feature extraction (without explicit ABCD scoring)**
  - Deep visual embeddings
  - Shape descriptors
  - Border statistics
  - Color distribution features
  - Size-related measurements

Output

- Classification logits and predicted lesion class
- Image-derived feature vectors
- Confidence scores and provenance metadata

Note: Explicit ABCDE values are **not computed at this stage** and are deferred to the IT-Fusion stage for multimodal clinical reasoning.

Stage 5: MedProc — Medical Information Processing

Purpose

To extract structured clinical information from patient-reported symptoms, medical history, and uploaded documents.

Input

- Chat responses describing symptoms
- Uploaded biopsy or blood test reports
- Prior lesion images (optional)

Preprocessing Required

- Text normalization
- Medical keyword extraction
- Parsing of numerical values (e.g., LDH levels)

Operations Performed

- Convert patient responses into clinical indicators (itching, inflammation, spreading)

- Derive **E – Evolution** using symptom progression and timeline reasoning
- Process non-dermatological symptoms for general medical support
- Construct structured clinical feature representations

Output

- Structured **Evolution (E)** value
- Symptom vectors
- Parsed medical findings
- JSON with reasoning for each extracted feature

Stage 6: Image + Text Fusion (IT-Fusion) — ABCDE Evaluation & Clinical Reasoning (Updated)

Purpose

To perform **final ABCDE evaluation, risk scoring, and overall medical interpretation** by integrating image-based features and textual clinical evidence.

Input

- Visual feature embeddings and geometric descriptors from HC

- Segmentation metadata from SLS and SLRC
- Structured clinical features and evolution data from MedProc
- Patient symptom summaries and timelines

Preprocessing Required

- Feature normalization and alignment
- Validation of lesion scale, mask quality, and temporal consistency
- Detection of missing or conflicting modality inputs

Operations Performed

ABCDE Evaluation (Final)

- **A – Asymmetry:** computed using shape imbalance and mask geometry
- **B – Border Irregularity:** derived from contour complexity metrics
- **C – Color Variation:** evaluated using color distribution and clustering features
- **D – Diameter:** estimated using lesion size with pixel-to-millimeter scaling

- **E – Evolution:** inferred from MedProc symptom timelines and reports

Risk Assessment & Medical Interpretation

- Fusion of ABCDE values with classification confidence
- Rule-based and learned risk calibration
- Identification of high-risk patterns requiring clinician review

JSON Construction

- Final lesion class
- Final ABCDE values
- Calibrated risk score
- Stage-wise explanations and evidence
- Flags for uncertainty or missing data
- Provenance metadata

Output

A **single unified clinical JSON** used across all downstream modules, including VQA.

Stage 7: Visual Question Answering (VQA)

Purpose

To interact with patients and clinicians and answer questions using fused clinical knowledge.

Input

- Unified JSON from IT-Fusion
- Patient or clinician queries

Operations Performed

- Identify relevant components of the unified JSON
- Query appropriate modules when required
- Apply reinforcement-based validation to reduce hallucinations
- Maintain conversational context

Output

- Accurate, context-aware responses
- Clinically grounded explanations
- Updated chat history

Stage 8: Frontend–Backend and API Layer

Purpose

To provide the user interface and backend orchestration.

Inputs: User actions, uploaded files, messages
Outputs: Real-time results, visualizations, and chat responses

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Stage 9: Testing and Deployment

Purpose

To ensure system reliability and accuracy.

Operations

- Model performance evaluation
- Integration and stress testing
- Performance benchmarking
- Deployment to production environment

End-to-End Flow Summary

1. Patient provides image and symptoms via chat
2. **SLS** → lesion segmentation mask
3. **SLRC** → clean color ROI

4. **HC** → classification + feature extraction
5. **MedProc** → structured evolution and clinical features
6. **IT-Fusion** → final ABCDE values + risk score + unified JSON
7. **VQA** → answers patient queries using clinical JSON
8. Frontend displays results
9. System is tested and deployed

Final Note (Design Justification)

In this architecture, **ABCDE metrics are computed using deterministic, clinically grounded algorithms**, while **machine learning models are used for classification, feature extraction, multimodal fusion, and risk interpretation**. This hybrid design maximizes **explainability, clinical validity, and system robustness**, making it well-suited for real-world dermatology decision support and M.Tech-level research.

Dataset

Stage Name	Dataset Name(s)	Training	Testing	Validation	Total

SLS (Skin Lesion Segmentation)	ISIC 2016 – Task 1 Segmentation	900	379	100	**1379**
SLS – Extended	ISIC 2017 – Task 1 Segmentation	2000	600	300	**2900**
SLRC (ROI Cropping)	ROI crops generated from ISIC 2016 + ISIC 2017 segmentation outputs	2900	979	400	**4279**
HC (Level 1 – Benign vs Malignant)	ISIC 2018 Classification Dataset (10,015 images)	7000	2000	1015	**10,015**
HC – Combined ISIC Stages	ISIC 2019 (25,331 images) + ISIC 2020 (33,126 images)	40,000	10,000	8,457	**58,457**
HC – Extended Multi-Class Dermatology	ISIC 2016 Task 3 + ISIC 2016 Task 3B + ISIC 2017 Classification + ISIC 2024 (~18k images)	20,000	8,000	2,900	**30,900**
HC – Large Clinical Dataset	BCN-20000 (20,000 clinical dermatology images)	14,000	4,000	2,000	**20,000**
HC – Dermoscopic Pigmented Lesions	MiLK-10K (10,015 dermoscopic images)	7000	2000	1015	**10,015**
MedProc (Medical Text Understanding)	MIMIC-III + MIMIC-IV Clinical Notes	50,000	10,000	5,000	**65,000**
IT Fusion (Image + Text Fusion Model)	Derm1M (1M images) + HAM10000 + PAD-UFES-20 + ISIC 2016-2024 + BCN-20000 + MIMIC-III/IV clinical text	700,000	200,000	100,000	**1,000,000 +**
VQA (Virtual Question Answering)	DermaVQA + SkinCancerVQA + SkinQA	80,000	15,000	10,000	**105,000**

	(clinical) + PathVQA + ISIC-Derived DermaQA				

Models

Stage name	**Model name**	**Paper name**	**Is code available**	**Paper link**	**Paper citation**	**Use of model**	**Advantage of model**	**Disadvantage of model**
SLS (Segmentation)	ARCUNet	ARCUNet: Enhancing Skin Lesion Segmentation with Residual Connections and Attention	Yes (NIH/PMC article typically links to code)	https://pmc.ncbi.nlm.nih.gov/articles/PMC11920274/		Dermatology-focused U-Net variant with residual and attention modules for skin lesion segmentation.	Designed specifically for skin lesions; strong boundary accuracy and easy to integrate into existing U-Net pipelines.	Not necessarily the absolute top performer compared to newest transformer hybrids; slightly heavier than plain U-Net.
HC (Classification)	EfficientFormerV2	EfficientFormerV2: A Lightweight Vision Transformer for Efficient Image Recognition	Yes (official GitHub)	https://github.com/snap-research/EfficientFormer		Backbone for benign vs malignant and multi-class lesion classification on dermoscopy/clinical crops.	Very efficient transformer backbone with good accuracy and low VRAM usage; suitable for multi-head HC on 12 GB GPU.	Not dermatology-specific; may slightly trail heavier Swin/ViT backbones on peak accuracy if you can afford them.

MedProc (medical text)	BioBERT	BioBERT: A Pre-trained Biomedical Language Representation Model for Biomedical Text Mining	Yes (Hugging Face, etc.)	https://sparknlp.org/2020/08/25/sent-biobert_clinical_base_cased.html		Text encoder for biopsy reports, blood test summaries, and medical descriptions.	Strong performance on biomedical NER and classification; widely used and supported.	Optimized mainly for literature-style biomedical text; not as strong on long, messy EHR notes as newer clinical models.
IT Fusion (handling missing modalities)	MARIA-style transformer	A Multimodal Transformer Model for Incomplete Healthcare Data (MARIA)	Likely yes (arXiv)	https://arxiv.org/abs/2412.14810		Fusion transformer that combines multiple modalities (image, labs, text, structured data) via masked attention to handle missing modalities.	Naturally handles incomplete modality sets, matching your real-world scenario where not all patients have all tests.	Requires custom integration of your encoders; not dermatology-specific; more engineering effort than simple concatenation.
VQA (baseline VLM)	R-LLaVA	R-LLaVA: Improving Med-VQA Understanding Through Visual Region of Interest Guidance	Yes (OpenReview/GitHub)	https://openreview.net/forum?id=HqdlX8yZJP		Med-VQA model that uses ROI information to guide visual attention and improve answers.	Aligns perfectly with your SLRC stage by using lesion ROIs; outperforms standard LLaVA-Med on multiple Med-VQA benchmarks.	General Med-VQA, not dermatology-specific; needs domain-specific fine-tuning and potentially model-size tuning for 12 GB GPU.

SLRC (Skin Lesion ROI Cropping)	OpenCV							
Frontend-Backend and API	React - Frontend , FastAPI- linking							

Literature Survey

Author Name	Year	Paper Title	Link	Difference from NutriDermAI	Insights Gained	Citations
Pan et al.	2025	A Multimodal Vision Foundation Model for Clinical Dermatology (PanDerm)	https://www.nature.com/articles/s41591-025-03747-y	Large dermatology foundation model across multiple imaging modalities; does not implement full conversational assistant, ABCDE workflow, or RL-based VQA like NutriDermAI.	Shows how self-supervised masked latent modeling and CLIP-style alignment on millions of dermatology images yields universal encoders reusable for SLS, HC, and IT Fusion.	Pan et al. (2025). A Multimodal Vision Foundation Model for Cli

						nical Dermatology (PanDerm) [https://www.nature.com/articles/s41591-025-03747-y]

Zeng et al.	2025	MM-Skin: Enhancing Dermatology Vision-Language Model with Expert-Curated Multimodal Dataset	https://arxiv.org/html/2505.06152v1	Builds dermatology VLM on expert-captioned multimodal data; lacks explicit MedProc rule engine, ABCDE metrics, and end-to-end chat deployment.	Demonstrates that expert-written long-form descriptions and multi-modality inputs significantly improve image–text understanding, guiding IT Fusion training and prompting strategies.	Zeng et al. (2025). MM-Skin: Enhancing Dermatology Vision-Language Model with Expert-Curated Multimodal Dataset [https://arxiv.org/html/250

						5.0 615 2v1]
Yan et al.	2 0 2 5	Derm1M: A Million-Scale Vision-Language Dataset Aligned with Hierarchical Dermatology Ontology	https://arxiv.org/abs/2503.14911	Provides 1M+ image–text pairs focused on pretraining; does not implement a staged assistant with SLS, ABCDE, and RL-based VQA.	Enables robust dermatology vision–language pretraining where DermLIP outperforms biomedical foundations in zero-shot classification, strongly informing IT Fusion backbones.	Yan et al. (20 25) . Der m1 M: A Mill ion- Sca le Visi on- Lan gua ge Dat ase t Alig ned wit h Hie rarc hic al Der ma tolo gy Ont olo gy [htt ps: //ar xiv. org

						/abs/2503.14911]
Aboulmira et al.	2025	Attention-Guided Multimodal Skin Disease Classification	https://www.sciencedirect.com/science/article/pii/S2352914825000887	Combines dermoscopic images and text in an attention-guided classifier; does not include conversational VQA, MedProc reasoning, or multi-stage ISIC/BCN integration.	Shows how attention-based fusion of image and text can improve diagnosis, suggesting architectural patterns for NutriDermAI's multimodal IT Fusion stage.	Aboulmira et al. (2025). Attention-Guided Multimodal Skin Disease Classification [https://www.sciencedirect.com/science/article

						e/pii/S2352914825000887]
Naseem et al.	2024	BioMedBLIP: Advancing Accuracy in Multimodal Medical Tasks Through Biomedically-Informed BLIP Models	https://medinform.jmir.org/2024/1/e56627	Targets radiology and other modalities; not specialized for dermatology or ABCDE-driven workflows.	Demonstrates that domain-adapted BLIP models outperform generic ones on medical tasks, informing model choice and pretraining strategy for IT Fusion and VQA.	Naseem et al. (2024). BioMedBLIP: Advancing Accuracy in Multimodal Medical Tasks Through Biomedically-Informe

						d BLIP Models [https://mediinform.jmir.org/2024/1/e56627]
ScaleFusionNet authors	2025	Transformer-Guided Multi-Scale Feature Fusion for Skin Lesion Segmentation	https://arxiv.org/html/2503.03327v3	Focuses on high-accuracy segmentation using Swin Transformers and deformable convolutions; does not implement ROI color reconstruction, ABCDE scoring, or multimodal reasoning.	Shows multi-scale fusion with Transformers significantly boosts Dice scores on ISIC, making such architectures strong SLS candidates feeding SLRC and downstream modules.	ScaleFusionNet authors (2025). Transformer-Guided Multi-Scale Feature Fusion

						for Skin Lesion Segmentation [https://arxiv.org/html/2503.03327v3]
ESC-UNet authors	2025	ESC-UNET: A Hybrid CNN and Swin Transformers for Skin Lesion Segmentation	https://www.sciencedirect.com/science/article/pii/S2666521225000614	Hybrid CNN–Swin model restricted to segmentation; not embedded in a full conversational assistant or VQA pipeline.	Confirms that combining CNN backbones with Swin Transformers yields high performance on ISIC data, supporting similar hybrid choices for NutriDermAI's SLS.	ESC-UNet authors (2025). ESC-UNET: A Hybrid CNN and Swin Transformerme

						rs for Skin Lesion Segmentation [https://www.sciencedirect.com/science/article/pii/S2666521225000614]
MRP-UNet authors	2025	Skin Lesion Segmentation with a Multiscale Input Fusion U-Net	https://www.nature.com/articles/s41598-025-92447-1	Improves segmentation boundaries via multiscale input fusion; does not support ROI reconstruction or text fusion as required in SLRC and IT Fusion.	Demonstrates that multiscale fusion and residual attention refine lesion borders and small-lesion sensitivity, directly benefiting SLS and ABCDE A/B estimation.	MRP-UNet authors (2025). Skin Lesion Seg

						mentation with a Multiscale Input Fusion U-Net [https://www.nature.com/articles/s41598-025-92447-1]
Meta-UNet authors	2025	Meta-UNet: Enhancing Skin-Lesion Segmentation with Multimodal Metadata	https://pmc.ncbi.nlm.nih.gov/articles/PMC12476315/	Uses metadata to refine segmentation masks; not connected to VQA, ABCDE rule logic, or conversational pipeline.	Shows that including metadata such as lesion location and patient info improves segmentation, encouraging early fusion of structured data in SLS and IT Fusion.	Meta-UNet authors (2025). Meta-UNet: Enh

						ancing Skin-Lesion Segmentation with Multimodal Metadata [https://pmc.ncbi.nlm.nih.gov/articles/PMC12476315/]
BiADATU-Net authors	2024	Intelligent Skin Lesion Segmentation Using Deformable Attention Transformer U-Net	https://onlinelibrary.wiley.com/doi/10.1111/srt.13783	Focuses purely on deformable-attention-based segmentation; does not integrate text or MedProc reasoning.	Highlights that deformable attention captures irregular lesion borders, improving boundary precision and supporting	BiADATU-Net authors (2024).

					robust ABCDE border feature extraction.	Intelligent Skin Lesion Segmentation Using Deformable Attention Transformer U-Net [https://onlinelibrary.wiley.com/doi/10.1111/srt.13783]

MUCM-Net authors	2024	MUCM-Net: A Mamba Powered UCM-Net for Skin Lesion Segmentation	https://www.explorationpub.com/Journals/em/Article/1001250	Provides lightweight segmentation; does not embed in a larger multimodal, conversational framework.	Delivers an efficient segmentation backbone suitable for 12GB GPUs, aligning with NutriDermAI's hardware constraints for the SLS stage.	MUCM-Net authors (2024). MUCM-Net: A Mamba Powered UCM-Net for Skin Lesion Segmentation [https://www.explorationpub.com/Journals/em/

						Article/1001250]
Hybrid dual-encoder authors	2024	A Hybrid Deep Learning Approach for Skin Lesion Segmentation Using Dual CNN–ViT Encoders	https://ieeexplore.ieee.org/iel8/6287639/10820123/10910106.pdf	Uses ResNet + ViT dual encoders for segmentation only, without downstream ROI reconstruction or multimodal heads.	Shows that dual-encoder CNN–ViT architectures with attention fusion improve segmentation, offering reusable patterns for SLS and feature extraction feeding IT Fusion.	Hybrid dual-encoder authors (2024). A Hybrid Deep Learning Approach for Skin Lesion Segmentation Using Dual CNN–ViT Encoders

						[https://ieeexplore.ieee.org/iel8/6287639/10820123/10910106.pdf]
Faggioli et al.	2025	Multimodal Learning for Skin Lesion Segmentation and Closed-Ended Visual Question Answering	https://www.dei.unipd.it/~faggioli/temp/clef2025/paper_197.pdf	Jointly tackles segmentation and closed-ended VQA for MEDIQA-MAGIC; does not implement full HC, MedProc, or dialogue management.	Confirms that shared representations between SLS and VQA benefit performance, supporting tight integration of SLS/SLRC with VQA in NutriDermAI.	Faggioli et al. (2025). Multimodal Learning for Skin Lesion Segmentation and Closed

						-Ended Visual Question Answering [https://www.dei.unipd.it/~faggioli/temp/clef2025/paper_197.pdf]
IReL team	2025	Tackling Multimodal Dermatology with CLIPSeg-Based Vision-Language Fusion	https://ceur-ws.org/Vol-4038/paper_208.pdf	Uses CLIPSeg-based segmentation and multimodal alignment for challenge tasks; lacks ABCDE scoring, rule-based MedProc, and RL answer selection.	Shows CLIP-style joint image–text representations help dermatology segmentation and VQA, informing CLIP-like encoders for IT Fusion and SLS.	IReL team (2025). Tackling Multimodal Derma

						tology with CLIPSeg-Based Vision-Language Fusion [https://ceur-ws.org/Vol-4038/paper_208.pdf]
Review authors	2024	A Review of U-Net-Based Deep Learning Models for Skin Lesion Segmentation	https://onlinelibrary.wiley.com/doi/10.1111/ima.70107	Survey of U-Net variants; does not implement a new pipeline or multimodal assistant.	Summarizes segmentation architectures and performance on standard datasets, guiding initial SLS backbone and baseline selection.	Review authors (2024). A Review of U-Net-Based Dee

						p Learning Models for Skin Lesion Segmentation [https://onlinelibrary.wiley.com/doi/10.1111/ima.70107]
Multimodal Skin Cancer Prediction authors	2025	Multimodal Skin Cancer Prediction: Integrating Dermoscopic Images and Multi-Modal Data	https://openbioinformaticsjournal.com/VOLUME/18/ELOCATOR/e18750362358444/	Integrates multiple data modalities for prediction but does not extend to conversational interaction, ABCDE rule logic, or VQA.	Shows benefits of fusing dermoscopic images with clinical data, supporting NutriDermAI's plan to combine HC outputs with MedProc in IT Fusion.	Multimodal Skin Cancer Prediction

						authors (2025). Multimodal Skin Cancer Prediction: Integrating Dermoscopic Images and Multi-Modal Data [https://openbioinformaticsjournal.com

						/VOLUME/18/ELOCATOR/e187503623584 44/]
Jütte et al.	2024	Integrating Generative AI with ABCDE Rule Analysis for Early Skin Cancer Detection	https://www.frontiersin.org/journals/medicine/articles/10.3389/fmed.2024.1445318/full	Uses generative AI plus ABCDE analysis for detection without building a multi-stage assistant with VQA, MedProc, and deployment stack.	Provides a working example of ABCDE-aware generative models for skin cancer, validating NutriDermAI's ABCDE-centric risk analysis design.	Jütte et al. (2024). Integrating Generative AI with ABCDE Rule Analysis for Early Skin Cance

						r Detection [https://www.frontiersin.org/journals/medicine/articles/10.3389/fmed.2024.1445318/full]
Chatterjee et al.	2021	Dermatological Expert System Implementing the ABCD Rule for Melanoma Detection	https://www.sciencedirect.com/science/article/abs/pii/S0957417420309337	Implements rule-based ABCD expert system using classical processing, not deep segmentation, multimodal fusion, or conversational flow.	Offers formalization of ABCD criteria and total dermoscopic scores as rules, directly informing MedProc's rule-based component.	Chatterjee et al. (2021). Dermatological Expert

						System Implementing the ABCD Rule for Melanoma Detection [https://www.sciencedirect.com/science/article/abs/pii/S0957417420309337]

Senan et al.	2021	Analysis of Dermoscopy Images by Using ABCD Rule for Melanoma Diagnosis	https://www.sciencedirect.com/science/article/pii/S2666285X21000017	Applies ABCD rule with classical methods; no VQA, LLMs, or multi-stage integration.	Reinforces clinical validity of ABCD rule and shows practical algorithms for automatic ABCD evaluation feeding HC and MedProc.	Senan et al. (2021). Analysis of Dermoscopy Images by Using ABCD Rule for Melanoma Diagnosis [https://www.sciencedirect.com/science/article/p

						ii/S266628 5X21000017]
Tas et al.	2022	Faster Evaluation of ABCD Rule and Total Dermoscopic Score for Nevomelanocytic Lesions	https://brieflands.com/journals/jssc/articles/102138	Optimizes ABCD/TDS computation speed; does not integrate into a multi-module conversational pipeline.	Demonstrates that ABCD/TDS evaluation can be accelerated for near real-time use, supporting responsive ABCDE scoring in interactive sessions.	Tas et al. (2022). Faster Evaluation of ABCD Rule and Total Dermoscopic Score for Nevomelanocytic Lesions [https://brieflan d

						s.com/journals/jssc/articles/102138]
Shetty et al.	2022	Skin Lesion Classification of Dermoscopic Images Using Hybrid CNN Models and ABCD Rule Features	https://www.nature.com/articles/s41598-022-22644-9	Combines CNN features with ABCD descriptors for classification but omits text reasoning, MedProc, and conversational frontends.	Validates the benefit of explicitly incorporating ABCD-based features into deep classifiers, informing HC feature engineering.	Shetty et al. (2022). Skin Lesion Classification of Dermoscopic Images Using Hybrid CNN Models and ABCD

						Rule Features [https://www.nature.com/articles/s41598-022-22644-9]
Mevorach et al.	2025	A Comparison of Skin Lesions' Diagnoses Between AI-Based Systems and Dermatologists Using ABCDE Rule	https://pmc.ncbi.nlm.nih.gov/articles/PMC12071753/	Evaluates existing AI systems vs dermatologists on ABCDE-based diagnosis; does not propose a new architecture.	Provides clinical evidence that ABCDE-based AI assessment can approach dermatologist performance, supporting NutriDermAI's ABCDE emphasis.	Mevorach et al. (2025). A Comparison of Skin Lesions' Diagnoses Between

						AI-Based Systems and Dermatologists Using ABCDE Rule [https://pmc.ncbi.nlm.nih.gov/articles/PMC12071753/]

Yim et al.	2024	DermaVQA: A Multilingual Visual Question Answering Dataset for Dermatology	https://dl.acm.org/doi/10.1007/978-3-031-72086-4_20	Releases dermatology VQA dataset and baselines but not a full multi-stage assistant with ABCDE, MedProc, and RL-based VQA.	Defines dermatology-specific VQA tasks and multilingual questions, guiding VQA training targets and evaluation for NutriDermAI.	Yim et al. (2024). DermaVQA: A Multilingual Visual Question Answering Dataset for Dermatology [https://dl.acm.org/doi/10.1007/978-3-031-72086-

						4_20]
Saeed et al.	2024	MediFact at MEDIQA-M3G 2024: Medical Question Answering with Weakly Supervised Answer Generation	https://aclanthology.org/2024.clinicalnlp-1.31/	Uses weak supervision for answer generation in multimodal medical tasks; lacks explicit ABCDE computation and full lesion pipeline.	Shows how weakly supervised and knowledge-augmented training can compensate for scarce labels in medical VQA, applicable to NutriDermAI's training regime.	Saeed et al. (2024). MediFact at MEDIQA-M3G 2024: Medical Question Answering with Weakly Supervised Answer Generation [https://a

						clanthology.org/2024.clinicalnlp-1.31/]
Kim et al.	2024	TEAM MIPAL at MEDIQA-M3G 2024: Large VQA Models for Multimodal Medical Answer Generation	https://aclanthology.org/2024.clinicalnlp-1.30.pdf	Adapts large VQA models for dermatology tasks but does not decompose pipeline into SLS, HC, MedProc, or use RL answer selection.	Provides fine-tuning and answer ranking strategies for large multimodal VQA, informing prompt and scoring design for NutriDermAI's VQA stage.	Kim et al. (2024). TEAM MIPAL at MEDIQA-M3G 2024: Large VQA Models for Multimodal Medical

						Answer Generation [https://aclanthology.org/2024.clinicalnlp-1.30.pdf]
H3N1 Lab	2025	DermKEM: A Knowledge-Enhanced Ensemble for Dermatology VQA	https://ceur-ws.org/Vol-4038/paper_202.pdf	Ensembles VLMs with external knowledge; does not tie in ABCDE scoring or structured MedProc outputs as knowledge sources.	Shows that knowledge-enhanced ensembles improve VQA accuracy, suggesting integration of MedProc's JSON reasoning into VQA as structured knowledge.	H3N1 Lab (2025). DermKEM: A Knowledge-Enhanced Ensemble for Dermatolo

						gy VQA [https://ceur-ws.org/Vol-4038/paper_202.pdf]
Madavan et al.	2025	Med-GRIM: Enhanced Zero-Shot Medical VQA Using Prompt Ensembles	https://openaccess.thecvf.com/content/ICCV2025W/MRR%2025/papers/Madavan_Med-GRIM_Enhanced_Zero-Shot_Medical_VQA_using_prompt-e.pdf	Targets zero-shot Med-VQA using prompt ensembles; not dermatology-specific and lacks RL-based training.	Demonstrates that combining prompts and VLMs improves zero-shot performance, inspiring ensemble and reranking strategies for NutriDermAI's VQA.	Madavan et al. (2025). Med-GRIM: Enhanced Zero-Shot Medical VQA Using Prompt Ens

						embles [https://openaccess.thecvf.com/content/ICCV2025W/MRR%202025/papers/Madavan_Med-GRIM_Enhanced_Zero-Shot_Medical_VQA_using

						_prompt-e.pdf]
Li et al.	2022	Medical Visual Question Answering via Corresponding Feature Fusion and Semantic Attention	https://www.aimspress.com/article/doi/10.3934/mbe.2022478	Designs semantic attention and feature fusion for Med-VQA on radiology, not dermatology.	Introduces corresponding feature fusion between image and question with semantic attention, useful for designing NutriDermAI's IT Fusion/VQA heads.	Li et al. (2022). Medical Visual Question Answering via Corresponding Feature Fusion and Semantic Attention [https://www.aimspr

						ess.com/article/doi/10.3934/mbe.2022478]
DSAP authors	2024	Bridging the Semantic Gap in Medical Visual Question Answering with Dynamic Semantic-Adaptive Prompting	https://ieeexplore.ieee.org/iel8/42/11230061/11039185.pdf	Uses adaptive prompting to improve Med-VQA; not tailored to dermatology or ABCDE.	Shows dynamic prompts can narrow the semantic gap between image and question, suggesting a prompt-adaptor layer for NutriDermAI's VQA.	DSAP authors (2024). Bridging the Semantic Gap in Medical Visual Question Answering with Dynami

						c Semantic-Adaptive Prompting [https://ieeexplore.ieee.org/iel8/42/1123006 1/11039185.pdf]
MedVLM-R1 authors	2025	MedVLM-R1: Incentivizing Medical Reasoning Capability with GRPO for VQA	https://papers.miccai.org/miccai-2025/paper/3267_paper.pdf	Applies GRPO RL to radiology VQA; not specifically for dermatology or ABCDE scoring.	Provides an RL framework with format and accuracy rewards for VQA, directly informing RL-based training for dermatology VQA in NutriDermAI.	MedVLM-R1 authors (2025). MedVLM-R1: Incentivizing

						Medical Reasoning Capability with GRPO for VQA [https://papers.miccai.org/miccai-2025/paper/3267_paper.pdf]
RL VQA Validation authors	2023	A Reinforcement Learning Approach for VQA Validation	https://www.sciencedirect.com/science/article/pii/S136184152300083X	Uses RL to validate and improve generic VQA answers; not focused on medical images.	Shows RL can select or filter better answers, supporting NutriDermAI’s use of RL to reduce hallucinations and improve reliability.	RL VQA Validation authors (2023)

						. A Reinforcement Learning Approach for VQA Validation [https://www.sciencedirect.com/science/article/pii/S136184152300083X]

MedThink authors	2025	MedThink: A Rationale-Guided Framework for Explaining Medical Visual Question Answering	https://aclanthology.org/2025.findings-naacl.415.pdf	Creates rationale-generating Med-VQA models; not dermatology-specific or ABCDE-aware.	Demonstrates how to generate textual rationales alongside answers, aligning with NutriDermAI's goal of JSON explanations and interpretable responses.	MedThink authors (2025). MedThink: A Rationale-Guided Framework for Explaining Medical Visual Question Answering [https://aclanthology.org/20

						25.findings-naacl.415.pdf]
HEALMQA authors	2024	HEALMQA: A Healthcare Multimodal Question Answering Benchmark	https://openreview.net/pdf/1d0a6992d0bc72959294fe86ec6553f6cbd3ee7a.pdf	Provides a broad healthcare multimodal QA benchmark; not dermatology or ABCDE-focused.	Offers benchmark design and evaluation ideas for multimodal QA, useful to structure NutriDermAI's test scenarios.	HEALMQA authors (2024). HEALMQA: A Healthcare Multimodal Question Answering Benchmark [https://openrevi

						ew.net/pdf/1d0a6992d0bc72959294fe86ec6553f6cbd3ee7a.pdf]
Lozano et al.	2025	Synthesizing High-Quality Visual Question Answering from Medical Literature	https://arxiv.org/html/2510.25867	Generates VQA samples from text; not focused on dermatology images or ABCDE.	Shows how to synthesize medically grounded VQA data, which can be adapted to generate extra dermatology QA pairs for training.	Lozano et al. (2025). Synthesizing High-Quality Visual Question Answering from Medic

						al Literature [https://arxiv.org/html/2510.25867]
Sviridov et al.	2025	Medical Multimodal Multi-agent Dialogue Benchmark	https://aclanthology.org/2025.emnlp-main.1353.pdf	Designs a multi-agent medical dialogue benchmark; not specific to dermatology or ABCDE rules.	Highlights multi-agent setups and metrics for diagnostic dialogue, offering patterns for NutriDermAI's dialogue and escalation design.	Sviridov et al. (2025). Medical Multimodal Multi-agent Dialogue Benchmark [https://aclantholo

						gy.org/2025.emnlp-main.1353.pdf]
Dermatology Chatbot authors	2024	Dermatology Chatbot: An AI-Driven Solution for Accessible Skin Care	https://www.americaspg.com/article/pdf/3773	Implements an image+text dermatology chatbot with Hybrid U-Net and MobileNet-V3 but without multimodal LLMs, ABCDE rules, or RL-based VQA.	Shows real-world implementation of dermatology chatbot integrating CNNs and rule-based logic, validating NutriDermAI's multi-stage assistant concept.	Dermatology Chatbot authors (2024). Dermatology Chatbot: An AI-Driven Solution for Accessible Skin Care

						[https://www.americaspg.com/article/pdf/3773]
Case study authors	2024	Design Considerations from a Wizard-of-Oz Dermatology Case Study with a Conversational Agent	https://dl.acm.org/doi/10.1145/3613905.3651891	Explores semi-human-in-the-loop dermatology conversational interface; not fully automated or multi-stage like NutriDermAI.	Identifies UX and safety considerations (e.g., disclaimers, escalation) for dermatology chatbots, guiding frontend and interaction design.	Case study authors (2024). Design Considerations from a Wizard-of-Oz Dermatology Case Study

						with a Conversational Agent [https://dl.acm.org/doi/10.1145/3613905.3651891]
AMIE authors	2025	Towards Conversational Diagnostic Artificial Intelligence (AMIE)	https://www.nature.com/articles/s41586-025-08866-7	Provides a general-purpose conversational diagnostic LLM; not dermatology- or ABCDE-specific.	Gives evaluation framework comparing AI vs clinicians and safety practices, which NutriDermAI can adapt for validation and deployment.	AMIE authors (2025). Towards Conversational Diagnostic Artifici

						al Intelligence (AMIE) [https://www.nature.com/articles/s41586-025-08866-7]
MIRA framework authors	2025	A Novel Framework for Fusing Modalities in Medical Retrieval-Augmented Generation	https://arxiv.org/html/2507.07902v1	General medical RAG fusion framework; not specifically dermatology or ABCDE-oriented.	Introduces attention-based modality fusion with query-dependent weighting, informing NutriDermAI's IT Fusion and retrieval design.	MIRA framework authors (2025). A Novel Framework for Fusing

						Modalities in Medical Retrieval-Augmented Generation [https://arxiv.org/html/2507.07902v1]
Transformer Attention Fusion authors	2025	Transformer Attention Fusion for Fine-Grained Medical Image Analysis	https://pmc.ncbi.nlm.nih.gov/articles/PMC12216456/	Applies dual attention mechanisms to other medical images (e.g., retinopathy); not dermatology-focused.	Shows that combining self- and cross-attention at multiple scales improves fine-grained recognition, relevant for detailed lesion representation.	Transformer Attention Fusion authors (2025).

						Transformer Attention Fusion for Fine-Grained Medical Image Analysis [https://pmc.ncbi.nlm.nih.gov/articles/PMC12216456/]

Attention Mechanism review authors	2023	Deep Learning Attention Mechanism in Medical Image Analysis	https://www.sciltp.com/journals/ijndi/articles/2504000011	Review of attention mechanisms across medical imaging tasks; no specific dermatology pipeline.	Synthesizes attention patterns that can be plugged into SLS, HC, and IT Fusion modules for NutriDermAI.	Attention Mechanism review authors (2023). Deep Learning Attention Mechanism in Medical Image Analysis [https://www.sciltp.com/journals/

						ijndi/articles/2504000011]
Swin Medical authors	2025	A Multi-Scale Attention-Based Swin Transformer Model for Medical Image Segmentation	https://www.nature.com/articles/s41598-025-22649-0	Generic Swin-based segmentation model; not dermatology-specific.	Validates Swin Transformers with multi-scale attention for segmentation, reinforcing their use in NutriDermAI's SLS.	Swin Medical authors (2025). A Multi-Scale Attention-Based Swin Transformer Model for Medical Image Segmenta

						tion [https://www.nature.com/articles/s41598-025-22649-0]
ISIC organizers	2016-2024	ISIC 2016–2024 Skin Lesion Segmentation and Classification Datasets	https://challenge.isic-archive.com/	Provide benchmark segmentation and classification datasets, but no multi-stage conversational system.	Supply core training and evaluation data for SLS and HC stages (including ISIC 2024 SLICE-3D extension) consistent with NutriDermAI's dataset plan.	ISIC organizers (2016-2024). ISIC 2016–2024 Skin Lesion Segmentation and Classificat

						ion Datasets [https://challenge.isic-archive.com/]
Tschandl et al.	2018	HAM10000: A Large Collection of Multi-Source Dermatoscopic Images	https://www.nature.com/articles/sdata2018161	Dermatoscopic dataset; does not include pipeline or multimodal reasoning.	Offers 10,015 labeled images across seven lesion types, supporting HC training and evaluation.	Tschandl et al. (2018). HAM10000: A Large Collection of Multi-Source Dermatoscopic Images [https:

						//www.nature.com/articles/sdata201816 1]
Combalia et al.	2019	BCN20000: Dermoscopic Lesions in the Wild	https://www.nature.com/articles/s41597-024-03387-w	Clinical and dermoscopy dataset; not coupled with VQA or ABCDE logic.	Provides large-scale clinical images that extend HC to real-world clinical photos beyond dermoscopy.	Combalia et al. (2019). BCN20000: Dermoscopic Lesions in the Wild [https://www.nature.com/articles/s

						41597-024-03387-w]
He et al.	2020	PathVQA: 30000+ Questions for Medical Visual Question Answering	https://arxiv.org/abs/2003.10286	Pathology VQA dataset; not dermatology-specific.	Pioneers medical VQA dataset design and evaluation metrics transferable to dermatology VQA design.	He et al. (2020). PathVQA: 30000+ Questions for Medical Visual Question Answering [https://arxiv.org/abs/2003.10286]

Rajkhowa et al.	2024	TM-PathVQA: 90000+ Textless Multilingual Questions for Pathology VQA	https://www.isca-archive.org/interspeech_2024/rajkhowa24_interspeech.pdf	Extends PathVQA to multilingual settings; not focused on skin.	Shows that multi-lingual medical VQA is feasible, suggesting multilingual goals for NutriDermAI.	Rajkhowa et al. (2024). TM-PathVQA: 90000+ Textless Multilingual Questions for Pathology VQA [https://www.isca-archive.org/interspeech_2024/rajk

						howa24_interspeech.pdf]
Yim et al.	2024	DermaVQA: Multilingual Visual Question Answering Dataset for Dermatology	https://papers.miccai.org/miccai-2024/211-Paper2444.html	Dermatology VQA dataset; does not define a full assistant system.	Directly supplies dermatology VQA data for training and evaluating NutriDermAI's VQA module.	Yim et al. (2024). DermaVQA: Multilingual Visual Question Answering Dataset for Dermatology [https://papers.miccai.org

						/miccai-2024/211-Paper2444.html]
Johnson et al.	2016	MIMIC-III Clinical Database v1.4	https://physionet.org/content/mimiciii/1.4/	ICU clinical text dataset; not dermatology-focused.	Provides large-scale clinical notes for pretraining MedProc's NLP models and reasoning components.	Johnson et al. (2016). MIMIC-III Clinical Database v1.4 [https://physionet.org/content/mimiciii/1.4/]

Hyland et al.	2023	MIMIC-IV-Note: Deidentified Free-Text Clinical Notes	https://physionet.org/content/mimic-iv-note/	Updated ICU text dataset; not specific to skin diseases.	Offers more recent clinical notes to improve general medical reasoning in MedProc.	Hyland et al. (2023). MIMIC-IV-Note: Deidentified Free-Text Clinical Notes [https://physionet.org/content/mimic-iv-note/]
Lau et al.	2018	VQA-RAD: Radiology Visual Question Answering Dataset	https://collab.dvb.bayern/spaces/TUMdlma/pages/73379920/	Radiology VQA dataset rather than dermatology.	Provides early Med-VQA design and question taxonomy useful for structuring dermatology VQA.	Lau et al. (2018). VQA-RAD:

						Radiology Visual Question Answering Dataset [https://collab.dvb.bayern/spaces/TUMdlma/pages/73379920/]
Lin et al.	2023	Medical Visual Question Answering: A Survey	https://www.sciencedirect.com/science/article/abs/pii/S0933365723001252	Survey of Med-VQA; no specific new model.	Summarizes methods and datasets for Med-VQA, informing NutriDermAI's VQA design space.	Lin et al. (2023). Medical Visual Questi

						on Answering: A Survey [https://www.sciencedirect.com/science/article/abs/pii/S0933365723001252]
Natural Language Processing of MIMIC authors	2019	Natural Language Processing of MIMIC-III Clinical Notes	https://arxiv.org/pdf/1912.12397.pdf	Applies NLP to clinical notes; not dermatology-specific.	Shows how to preprocess and model clinical free-text for entity and concept extraction, informing MedProc's pipeline.	Natural Language Processing of MIMIC authors

						(2019). Natural Language Processing of MIMIC-III Clinical Notes [https://arxiv.org/pdf/1912.12397.pdf]
Clinical Text Review authors	2020	Clinical Text Data in Machine Learning: Systematic Review	https://pmc.ncbi.nlm.nih.gov/articles/PMC7157505/	Systematic review; no new model.	Outlines best practices and challenges for clinical NLP, guiding MedProc design.	Clinical Text Review authors (2020). Clin

						ical Text Data in Machine Learning: Systematic Review [https://pmc.ncbi.nlm.nih.gov/articles/PMC7157505/]
ChatGPT Dermatology review authors	2025	The Role of ChatGPT in Dermatology Diagnostics	https://pmc.ncbi.nlm.nih.gov/articles/PMC12191462/	Review of LLM usage; not a specific deployed system like NutriDermAI.	Highlights strengths, risks, and limitations of LLMs in dermatology, supporting safe integration strategies.	ChatGPT Dermatology review authors (20

						25). The Role of ChatGPT in Dermatology Diagnostics [https://pmc.ncbi.nlm.nih.gov/articles/PMC12191462/]
Conversational Agents Effectiveness authors	2025	Effectiveness of AI-Driven Conversational Agents in Improving Dermatological Patient Engagement	https://www.jmir.org/2025/1/e69639/	Evaluates chat agents; does not propose NutriDermAI-like pipeline.	Provides evidence that conversational agents improve patient engagement, supporting NutriDermAI's chat interface.	Conversational Agents Effectiveness

						authors (2025). Effectiveness of AI-Driven Conversational Agents in Improving Dermatological Patient Engagement [https://www.jmir.org/2025/1/e696

						39/]
Medical Multimodal LLM review authors	2024	Medical Multimodal Large Language Models: A Systematic Review	https://www.sciencedirect.com/science/article/pii/S2950261625000512	Review of MMLLMs; not dermatology-specific.	Summarizes current multimodal LLM architectures and limitations, helping position NutriDermAI relative to state of the art.	Medical Multimodal LLM review authors (2024). Medical Multimodal Large Language Models: A Systematic Review [https://www.scienc

						edirect.com/science/article/pii/S2950261625000512]
MedVLThinker authors	2025	MedVLThinker: Simple Baselines for Multimodal Medical Reasoning	https://arxiv.org/html/2508.02669v2	Proposes simple baselines for multimodal reasoning; not dermatology-focused.	Shows that carefully designed simple multimodal baselines can perform competitively, suggesting efficient designs suited to 12GB hardware.	MedVLThinker authors (2025). MedVLThinker: Simple Baselines for Multimodal Medical Rea

						soning [https://arxiv.org/html/2508.02669v2]
Validation Dermatology MLLM authors	2025	Validation of a Dermatology-Focused Multimodal Large Language Model	https://pmc.ncbi.nlm.nih.gov/articles/PMC12608998/	Validates a dermatology MLLM clinically; does not describe a full multi-stage assistant pipeline.	Demonstrates clinical benefit of domain-specific multimodal models, supporting NutriDermAI's specialization strategy.	Validation Dermatology MLLM authors (2025). Validation of a Dermatology-Focused Multimodal Large

						Language Model [https://pmc.ncbi.nlm.nih.gov/articles/PMC12608998/]
ImageCLEF MEDIQA-MAGIC organizers	2025	ImageCLEFmed MEDIQA-MAGIC: Multimodal Medical Dialogue and Generation Benchmark	https://www.imageclef.org/2025/medical/mediqa	Benchmark challenge; not a specific assistant.	Provides task definitions and metrics for multimodal medical dialogue and QA that NutriDermAI can use for evaluation.	ImageCLEF MEDIQA-MAGIC organizers (2025). ImageCLEFmed MEDIQ

						A-MAGIC: Multimodal Medical Dialogue and Generation Benchmark [https://www.imageclef.org/2025/medical/mediqa]

WoundcareVQA authors	2025	WoundcareVQA: A Multilingual Visual Question Answering Dataset for Wound Care	https://www.sciencedirect.com/science/article/pii/S1532046425001170	Wound care VQA dataset; adjacent but not dermatology-specific.	Illustrates the construction of skin-related multilingual VQA datasets, providing patterns transferable to dermatology.	WoundcareVQA authors (2025). WoundcareVQA: A Multilingual Visual Question Answering Dataset for Wound Care [https://www.sciencedirect.co

						m/science/article/pii/S1532046425001170]
Skin Cancer Classification Perspectives authors	2024	Decoding Skin Cancer Classification: Perspectives, Insights, and Future Directions	https://www.nature.com/articles/s41598-024-81961-3	Perspectives review; not an implementation of a pipeline.	Provides technical and clinical perspectives on skin cancer AI, informing HC design and evaluation choices.	Skin Cancer Classification Perspectives authors (2024). Decoding Skin Cancer Classification: Perspe

						ctives, Insights, and Future Directions [https://www.nature.com/articles/s41598-024-81961-3]